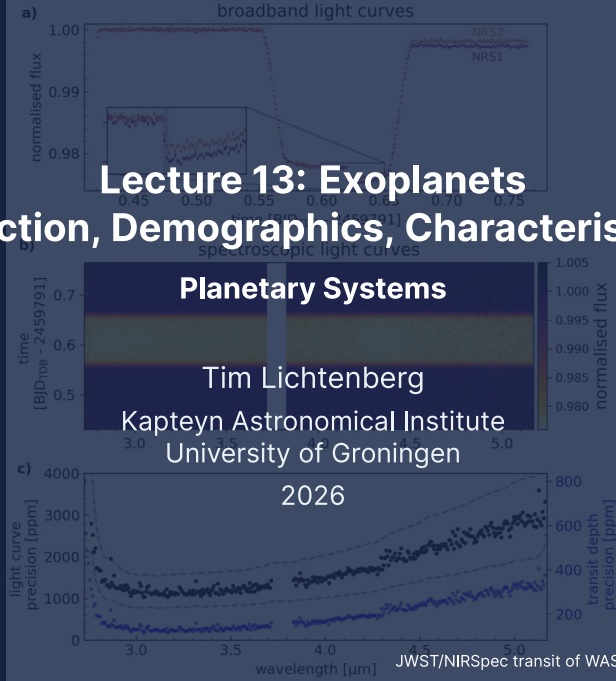


Lecture 13: Exoplanets

Detection, Demographics, Characterisation



Planetary Systems

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University of Groningen

Learning Objectives

By the end of this lecture, you will be able to:

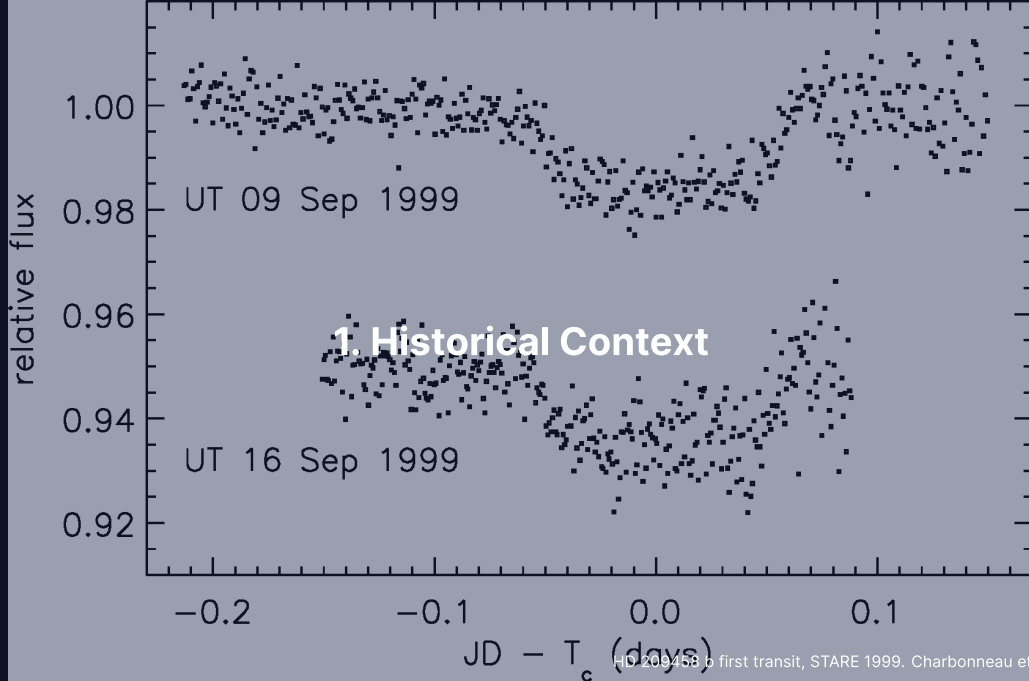
- Describe the main exoplanet detection methods and their biases
- Interpret the period-radius diagram and the radius valley
- Apply transit and radial-velocity geometry to derive bulk density
- Evaluate JWST-era atmospheric results and biosignature claims

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

- Lectures 9–11 surveyed the geology, atmospheres, and interiors of the solar planets
- Lecture 12 decoded small bodies and meteorites as planet formation fossils
- This lecture turns outward to thousands of exoplanets to test if these patterns are universal

Today: How do we detect, weigh, and characterise worlds beyond our solar system?



1. Historical Context

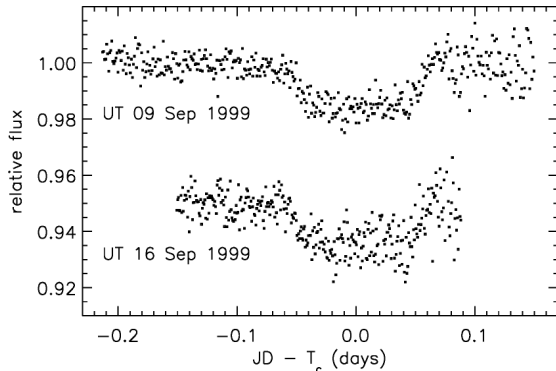
HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

From Zero to Six Thousand in Three Decades

Three decades: from the first detection to a census of over 6000 confirmed worlds.

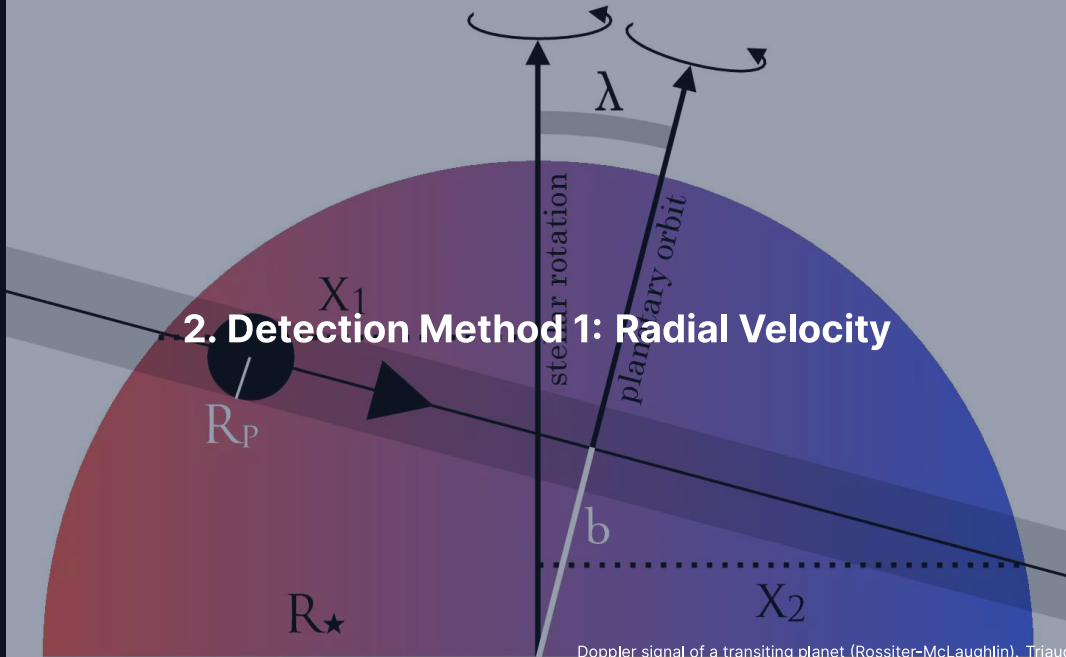
- **1992 (PSR B1257+12):** first exoplanets, around a pulsar, by timing (Wolszczan & Frail)
- **1995 (51 Pegasi b):** first planet around a Sun-like star; a 4.23-day orbit forced migration theory (Mayor & Queloz)
- **1999 (HD 209458 b):** first transit of an RV planet: a gas-giant radius and bulk density

HD 209458 b: First Ground-Based Transit (1999)



- STARE photometer, two nights in Sep 1999
- Successive transits dropped flux by $\sim 1.7\%$
- Independent detection by Henry et al. (2000)
- Exoplanets became physical objects with measurable radii

2. Detection Method 1: Radial Velocity



Radial Velocity: Doppler Reflex of the Star

- Star and planet orbit a common barycentre: $a_{\star} = (m_p/M_{\star}) a_p$
- Periodic Doppler shift of stellar absorption lines yields reflex velocity
- Semi-amplitude $K_{\star}$ (edge-on orbit; set $e = 0$ for a circular one):

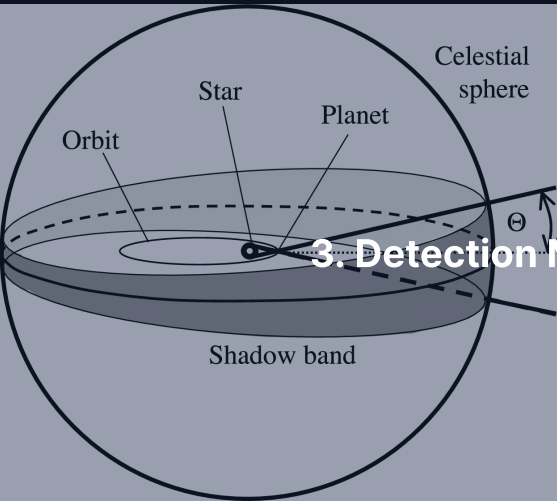
$$K_{\star} = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

- Jupiter at 5.2 AU: $K_{\star} \approx 12.5 \text{ m s}^{-1}$
- Earth at 1 AU: $K_{\star} \approx 9 \text{ cm s}^{-1}$

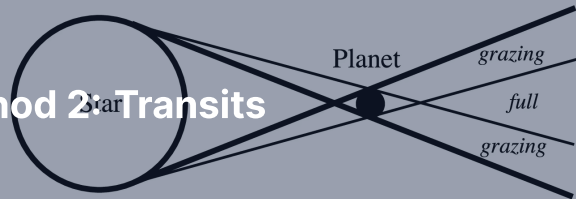
Noise Floor and the $m \sin i$ Degeneracy

RV measures only minimum mass $m_p \sin i$, with precision now limited by stellar jitter rather than spectrographs.

- **Noise floor:** ESPRESSO reaches 10 cm s^{-1} , but stellar granulation and spots produce $0.1\text{--}1 \text{ m s}^{-1}$ jitter
- **Degeneracy:** Orbit inclination i is unknown, so RV alone yields only a lower limit $m_p \sin i$
- **Breaking degeneracy:** Transits guarantee $\sin i \approx 1$, collapsing the ambiguity to yield the true planet mass

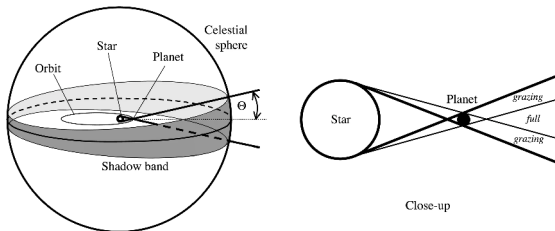


3. Detection Method 2: Transits



Close-up

Transit Geometry



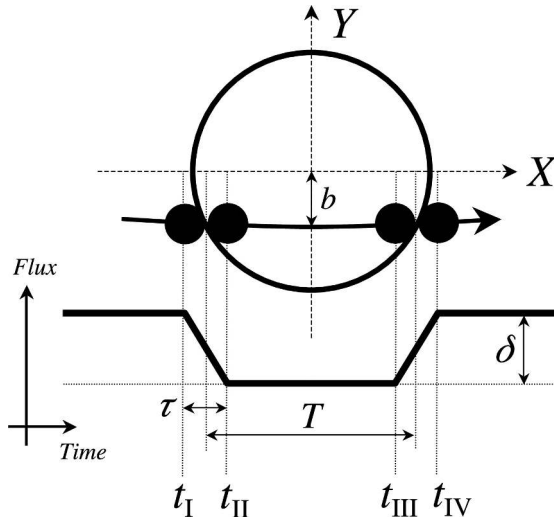
- Orbital plane near line of sight
→ planet crosses stellar disk
- Transit depth: $\delta = (R_p/R_\star)^2$
- Geometric probability:
 $p \approx R_\star/a$
- Strong selection bias toward short periods and large $R_p/R_\star$

Quick Scales

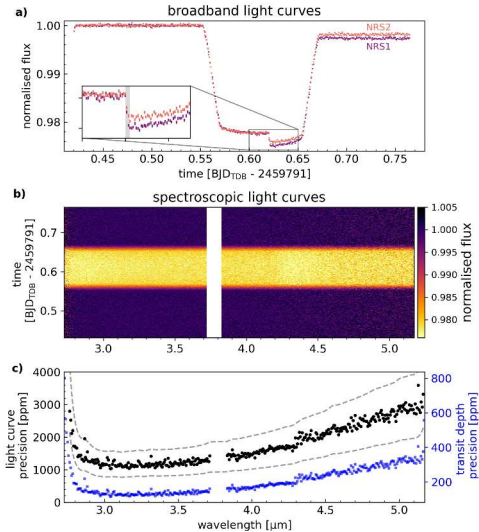
Transit depth scales inversely with the square of stellar radius, making small planets around small stars far easier to detect.

- **Jupiter around Sun:** $\delta \approx 1.01\%$, easily detected from ground-based telescopes
- **Earth around Sun:** $\delta \approx 84$ ppm, requires ultra-precise space photometry
- **Earth around M dwarf ($0.15 R_{\odot}$):** $\delta \approx 4000$ ppm, accessible from ground and space

Light Curve Anatomy

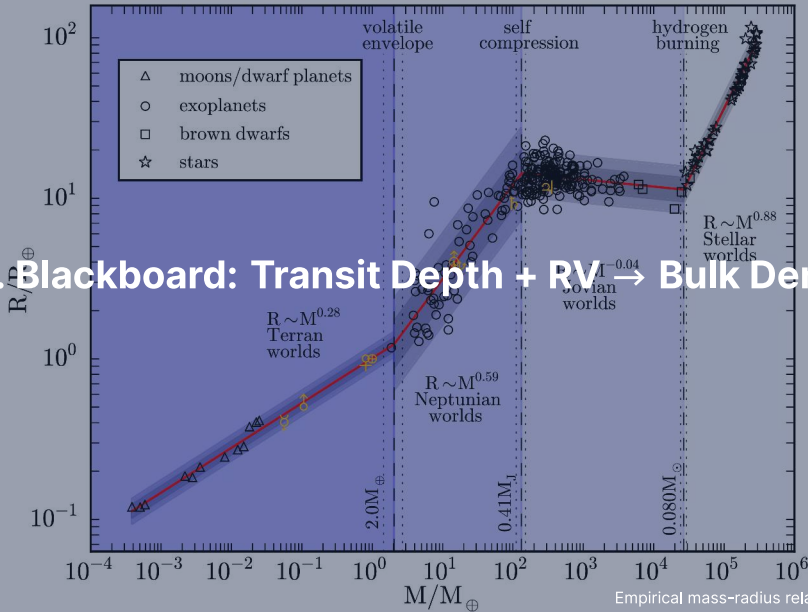


- Four contact times: t_I , t_{II} , t_{III} , t_{IV}
- Duration yields impact parameter b and orbital inclination
- Ingress and egress shape constrains stellar limb darkening
- Transit depth determines planet-to-star radius ratio $R_p/R_\star$



JWST/NIRSpec transit photometry of WASP-39 b: transit depth measured to 200–600 ppm per spectroscopic bin, pushing photometric noise below typical exoplanet absorption features. Alderson et al. (2023).

4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



Why Combine Transit and RV?

- Transit alone: yields planet radius R_p , but no mass
- RV alone: yields minimum mass $m_p \sin i$, but no radius
- Combined: **both** for the same planet $\rightarrow$ bulk density $\bar{\rho}_p$
- Bulk density turns an astronomical detection into a physical, geological body

To the board.

Result: Depth, Semi-Amplitude, Density

Transit depth $\delta = (R_p/R_\star)^2$ gives R_p ; the reflex velocity

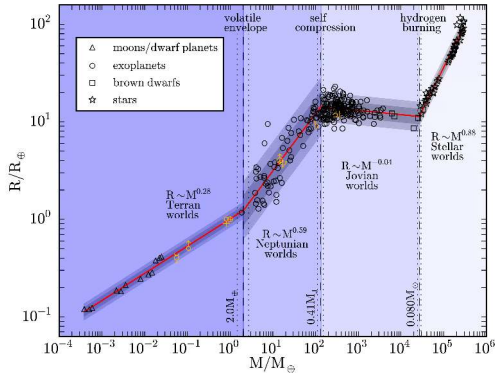
$K_\star = (2\pi G/P)^{1/3} m_p \sin i M_\star^{-2/3}$ gives $m_p \sin i$ (broad).

- A transit forces $\sin i \approx 1$ and collapses the $m_p \sin i$ degeneracy
- Both R_p and m_p measured for the same object
- Bulk density follows from elementary geometry:

$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

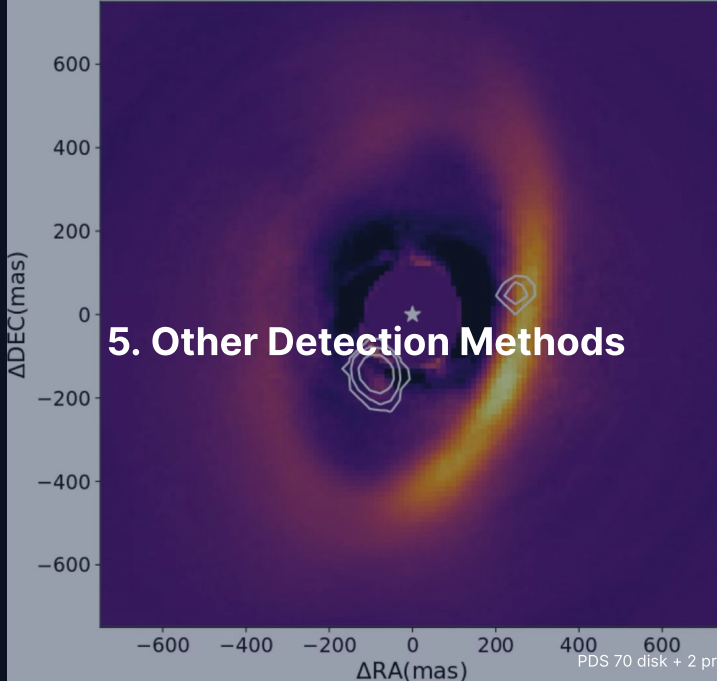
Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.

Density Maps to Composition



- $\bar{\rho} \approx 5.5 \text{ g cm}^{-3}$: rocky terrestrial (Earth, Venus)
- $\bar{\rho} \approx 1.3 \text{ g cm}^{-3}$: gas giant (Jupiter)
- $\bar{\rho} \approx 0.5 \text{ g cm}^{-3}$: inflated volatile envelope
- $\bar{\rho} \approx 2\text{--}4 \text{ g cm}^{-3}$: sub-Neptunes, unseen in the solar system

5. Other Detection Methods

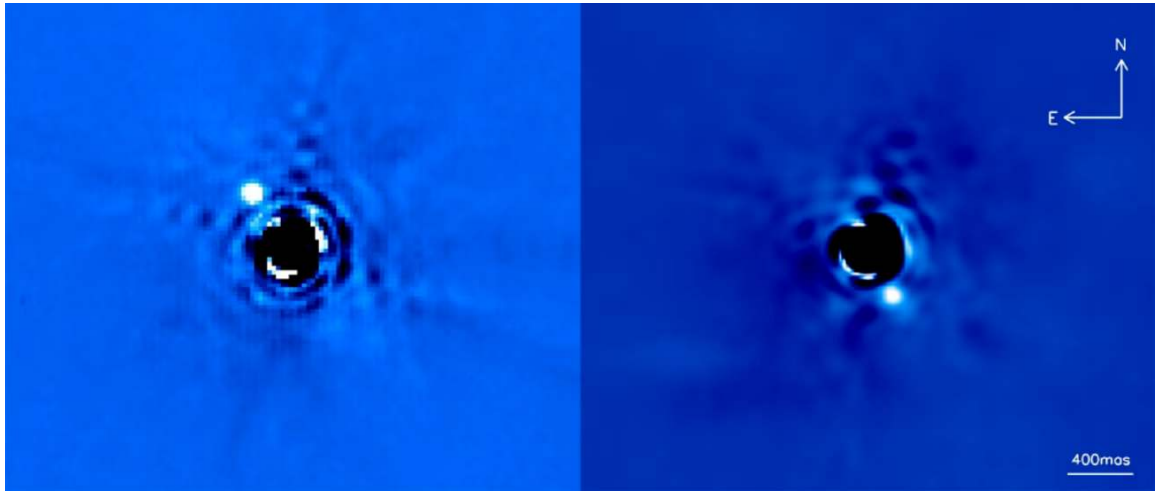


PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

Direct Imaging: Spatial Separation of Photons

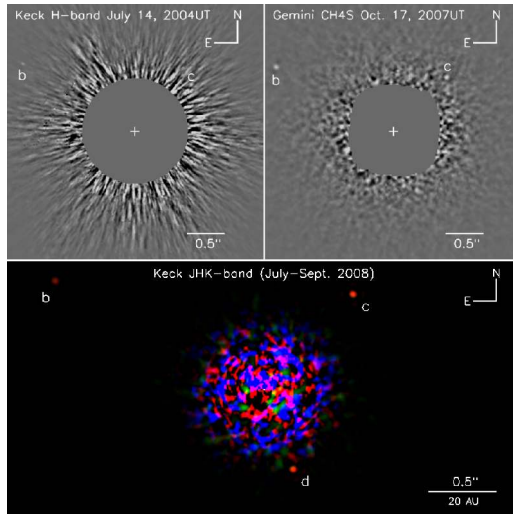
Direct imaging separates planet photons from the stellar glare.

- **Contrast:** suppress the star by 10^{-9} (Jupiter) to 10^{-10} (Earth twin) at separation $\theta = a/d$
- **Technology:** extreme adaptive optics plus coronagraphs against diffraction and speckle noise
- **Bias:** young, massive, self-luminous giants on wide orbits ($a > 10$ AU)

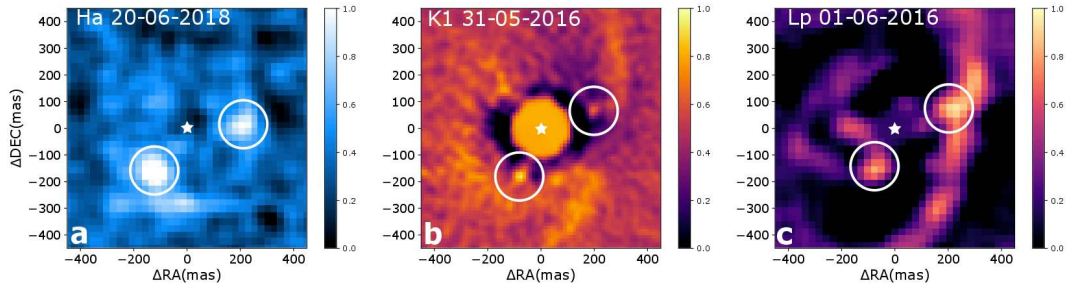


Direct image of β Pictoris b from Lagrange et al. (2010), showing orbital motion between 2003 and 2009 that confirms a bound $\sim 9 M_J$ companion at ~ 9 AU. Credit: ESO/A.-M. Lagrange et al.

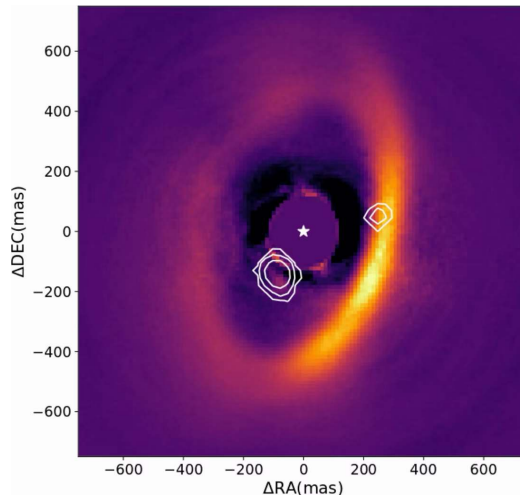
HR 8799: Four Imaged Giants



- Keck and Gemini AO discovery of three bound co-moving companions (Marois et al. 2008); planet e added in 2010
- Four planets confirmed across multiple epochs
- Masses $5\text{--}10 M_J$ at orbital separations 14–70 AU
- Young host star (30–60 Myr): the planets still glow from their contraction heat



PDS 70 b and c inside the disk gap in three bands: MUSE H α (a), SPHERE K1 (b), NACO L' (c); both protoplanets are actively accreting within the cleared cavity. Haffert et al. (2019).



Composite scattered-light image of the PDS 70 protoplanetary disk showing the parent circumstellar disk and two embedded protoplanets inside the cleared gap. Credit: Haffert et al. (2019).

Astrometry: Gaia Will Find Jupiter Analogues

Astrometry follows the star's reflex wobble on the sky and gives true masses.

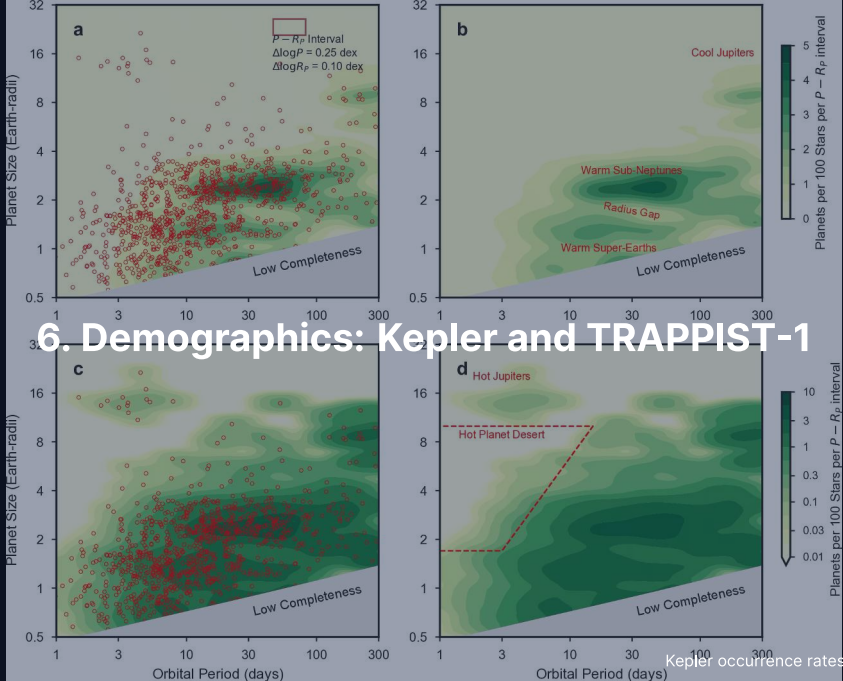
$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- **Signal:** a Jupiter analogue at 10 pc gives $\alpha \sim 500 \mu\text{as}$; an Earth analogue $\sim 0.3 \mu\text{as}$
- **Gaia DR4 (2026):** thousands of cold giants expected; $\sim 10 \mu\text{as}$ final precision
- **True mass:** the inclination i is measured, so no $m \sin i$ degeneracy

Microlensing, Timing, and Detection Biases

Each method covers its own part of parameter space, so the archive carries the biases of the methods.

- **Microlensing (Roman):** cool planets at 1–10 AU and unbound rogue worlds
- **Timing variations (TTVs):** dynamical masses for faint hosts such as TRAPPIST-1
- **Together:** transits and RV probe close-in orbits; imaging, astrometry and microlensing the wide ones

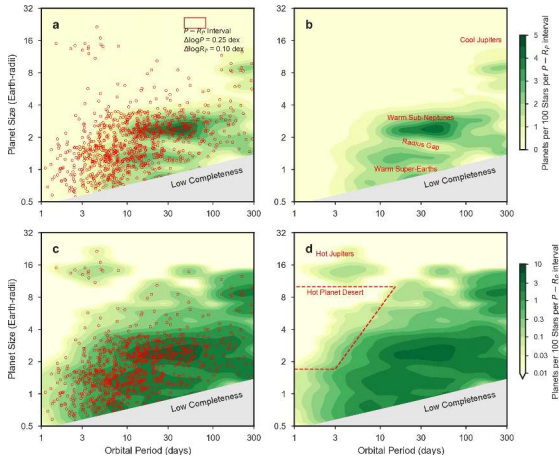


The Kepler Revolution

Kepler's computable detection efficiency delivered the first bias-corrected occurrence rates: planets outnumber stars in the Milky Way.

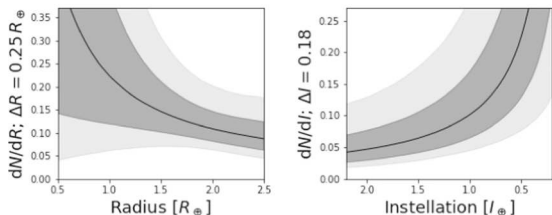
- **Planets are ubiquitous:** Average of > 1 planet per main-sequence star (Petigura et al. 2018)
- **Small planets dominate:** Super-Earths and sub-Neptunes are common; hot Jupiters are rare (0.5–1%)
- **Habitable terrestrial rate:** Occurrence of Earth-sized planets in the habitable zone is $\eta_{\oplus} \sim 0.4$ (Bryson et al. 2021)

Period-Size Occurrence Rate



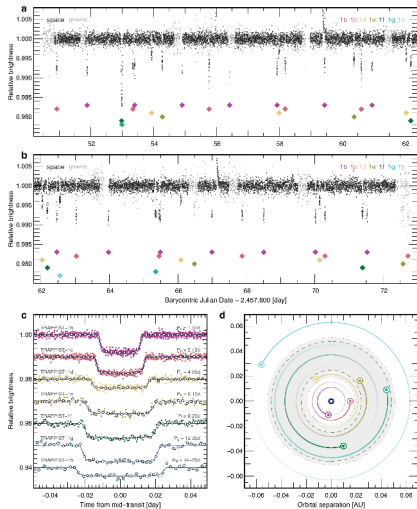
- California-Kepler Survey, after stellar parameter refinement
- Small planets common across all orbital periods
- Small planets outnumber giants at every period
- Fall-off beyond ~ 100 days is partly completeness-limited

$\eta_{\oplus}$: Earths in Habitable Zones



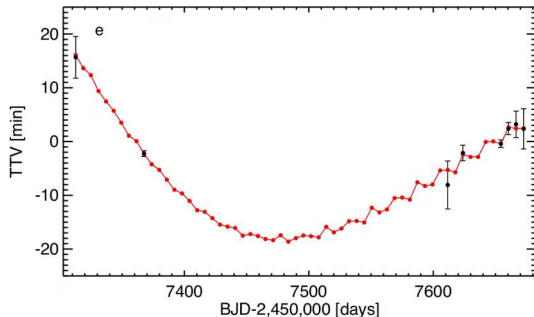
- Bryson et al. (2021) occurrence rate for Sun-like stars
- Marginalised over planet radius and stellar instellation
- Shaded bands represent 68% and 95% credible intervals
- Central estimate $\eta_{\oplus} \sim 0.4$: terrestrial planets in habitable zones are common

TRAPPIST-1: Seven Earths in Resonance



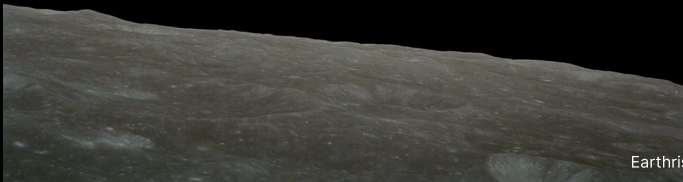
- M8 ultra-cool dwarf at 12 pc hosting seven Earth-sized worlds within 0.06 AU
- Resonant Laplace-like chain is consistent with early inward disk migration (Gillon et al. 2017)
- Three planets (e, f, g) orbit in the temperate habitable zone (Gillon et al. 2017)

TRAPPIST-1 Masses from TTVs

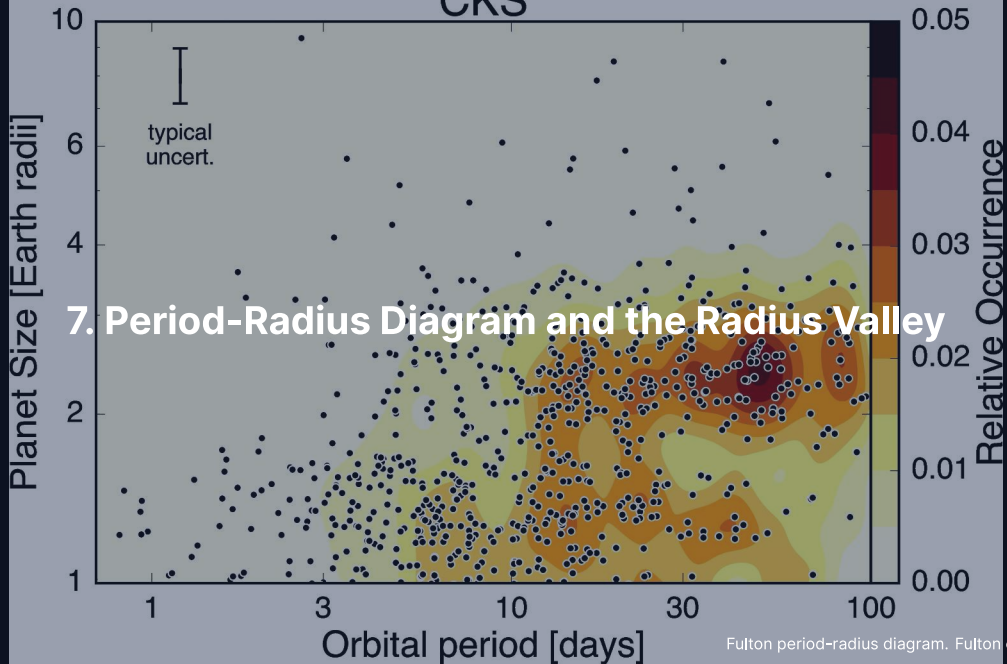


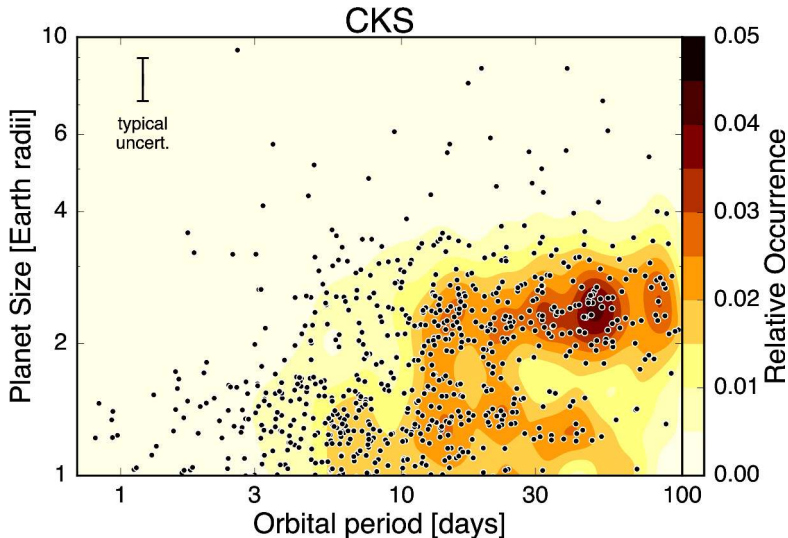
- Multi-planet transit-timing variations reach tens-of-minutes amplitude
- Simultaneous N-body dynamical modeling breaks mass degeneracies
- Determines individual masses and densities without requiring radial velocities

Break



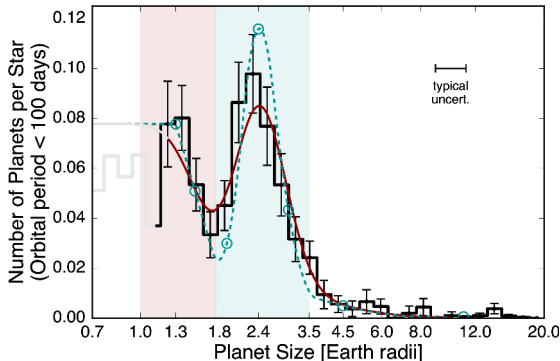
CKS





Period-radius distribution of small Kepler planets after stellar parameter refinement, revealing super-Earth and sub-Neptune populations separated by a deficit at $\sim 1.8 R_{\oplus}$. Fulton et al. (2017).

The Radius Valley (Fulton Gap)



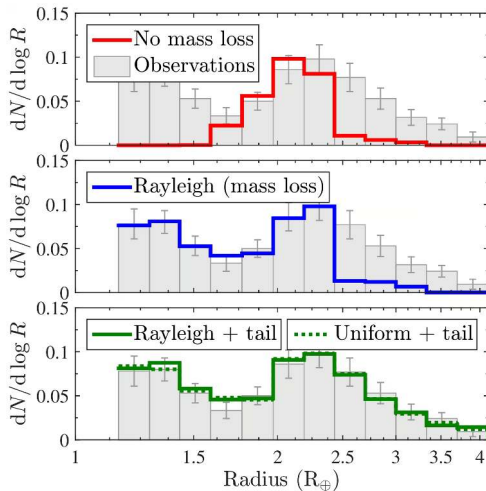
- Statistically significant deficit at $1.5\text{--}2.0 R_{\oplus}$
- Super-Earth population peaks at $\sim 1.3 R_{\oplus}$ (bare rocky cores)
- Sub-Neptune population peaks at $\sim 2.4 R_{\oplus}$ (volatile envelopes)
- Two stripping mechanisms: photoevaporation (stellar XUV) and core-powered mass loss (interior heat, no XUV needed)

Photoevaporation Strips H/He Envelopes

Energy-limited hydrodynamic escape driven by high stellar X-ray and EUV irradiation:

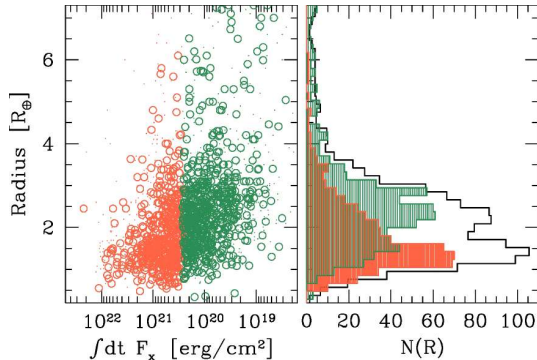
$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

- Young stars saturate XUV flux at $\sim 10^3 \times$ modern levels for ~ 100 Myr
- Strips light primordial H/He envelopes from lower-mass cores at close-in orbits
- Owen & Wu (2013): naturally carves the bimodal radius valley

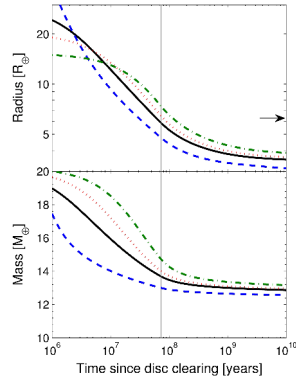


Core-powered mass loss model of the radius valley: envelope loss driven by heat from the cooling rocky interior produces a bimodal distribution with a gap at $\sim 1.8 R_{\oplus}$. Credit: Ginzburg et al. (2018).

Photoevaporation Model + Mass-Loss Tracks

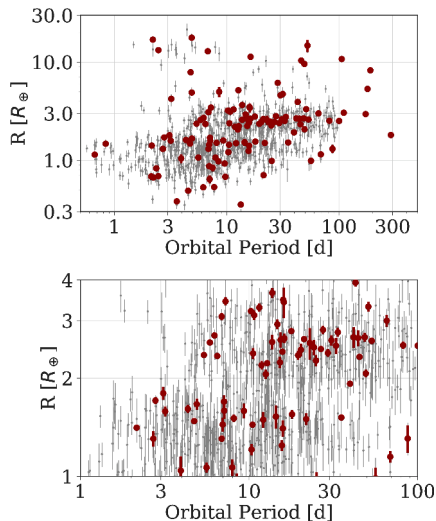


Left: predicted radius valley



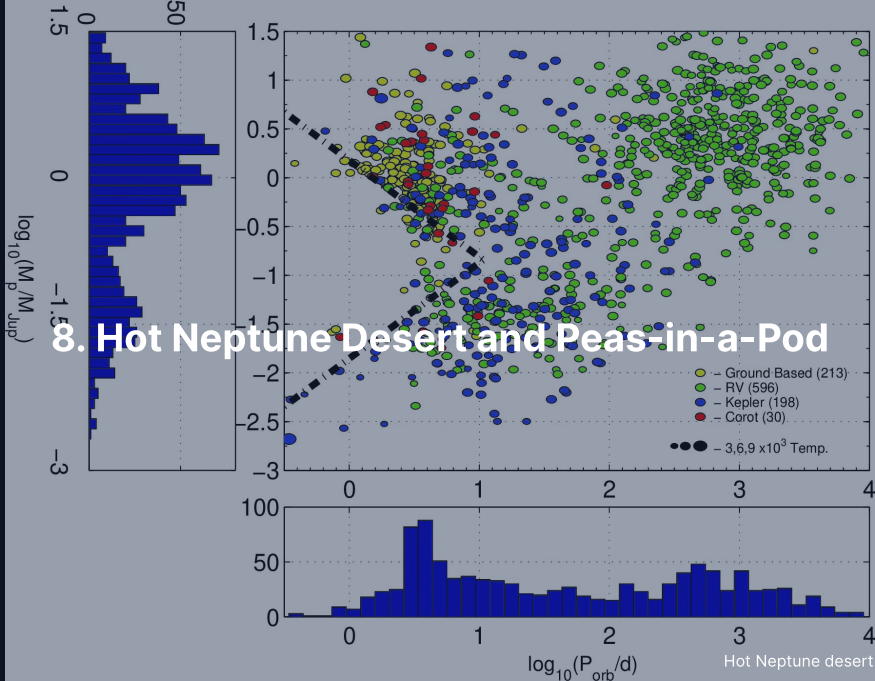
Right: time evolution of radius and mass; saturated XUV phase drives bimodal outcome (Owen & Wu 2013).

Valley Slope and Stripped Cores

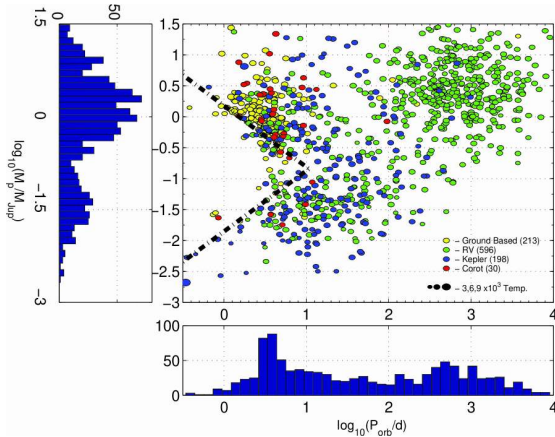


Credit: Van Eylen et al. (2018)

- Asteroseismic radii reveal the valley slope shifts to smaller R_p at longer periods
- Consistent with photoevaporation and core-powered cooling models
- Many close-in super-Earths are stripped cores rather than primordial rocky worlds



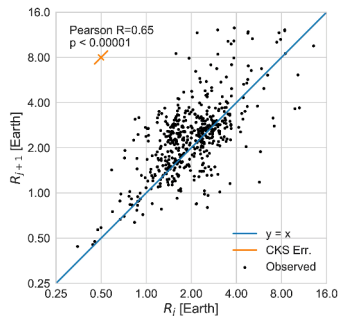
The Hot Neptune Desert



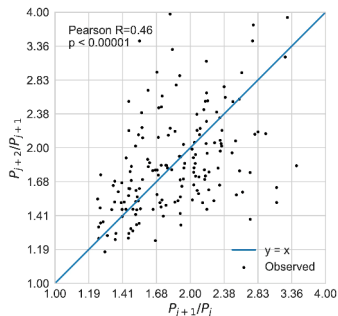
Credit: Mazeh et al. (2016)

- Pronounced deficit of short-period planets ($P < 5$ d) in the $10\text{--}100 M_{\oplus}$ mass range
- **Bottom edge:** Photoevaporation strips low-mass envelopes into super-Earths
- **Top edge:** Tidal Roche-lobe overflow strips inflated gas giants

Peas-in-a-Pod Architectures



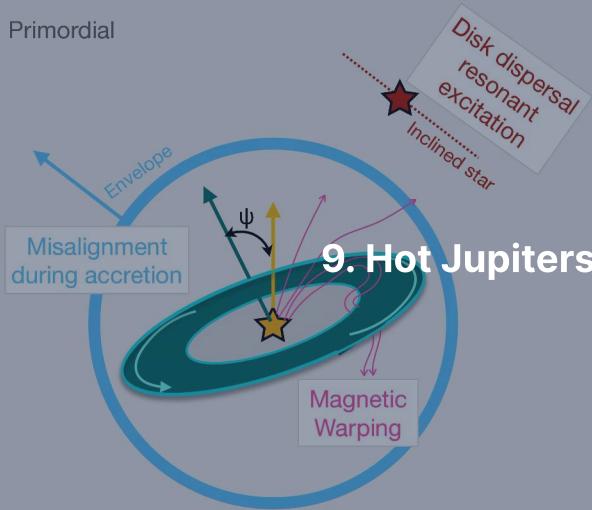
Adjacent radii: Pearson $r = 0.65$



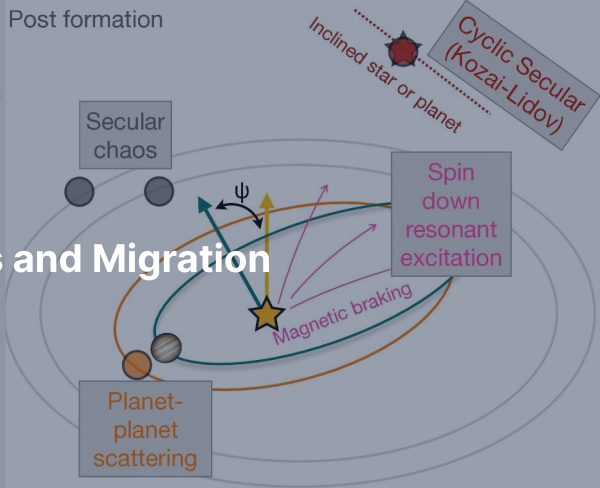
Adjacent period ratios: clustered

- Planets in compact multi-systems tend to have similar sizes and uniform orbital spacing (Weiss et al. 2018)
- Favours quiet in-situ growth or migration over chaotic giant impacts

Primordial

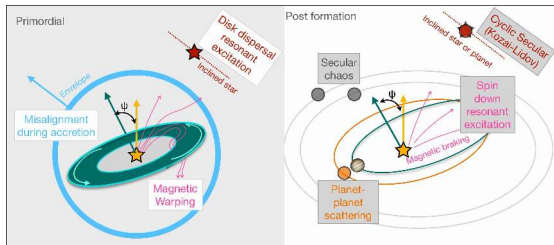


Post formation



9. Hot Jupiters and Migration

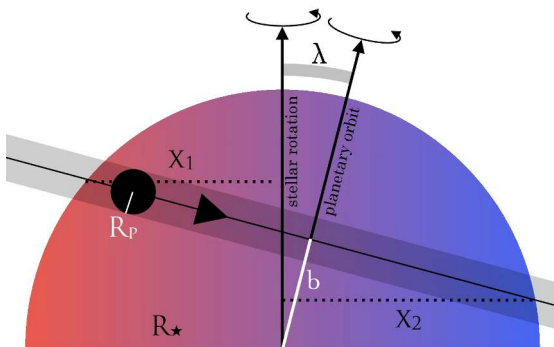
Hot Jupiter Migration Channels



Credit: Albrecht et al. (2022)

- **Disk migration (Type II):** disk torques move the planet inward smoothly, preserving low obliquity (in situ growth at 0.05 AU fails: too little solid mass, too hot)
- **High-eccentricity (Kozai-Lidov):** distant perturbers pump e ; tidal circularisation leaves wide misalignments
- **Planet scattering:** instability ejects giants and scrambles alignments

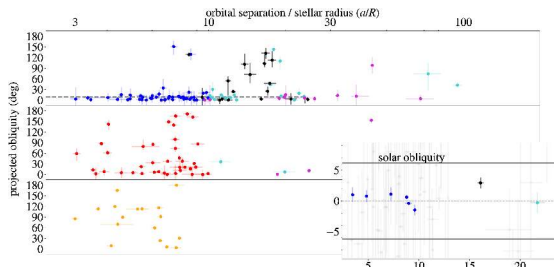
Rossiter-McLaughlin Effect



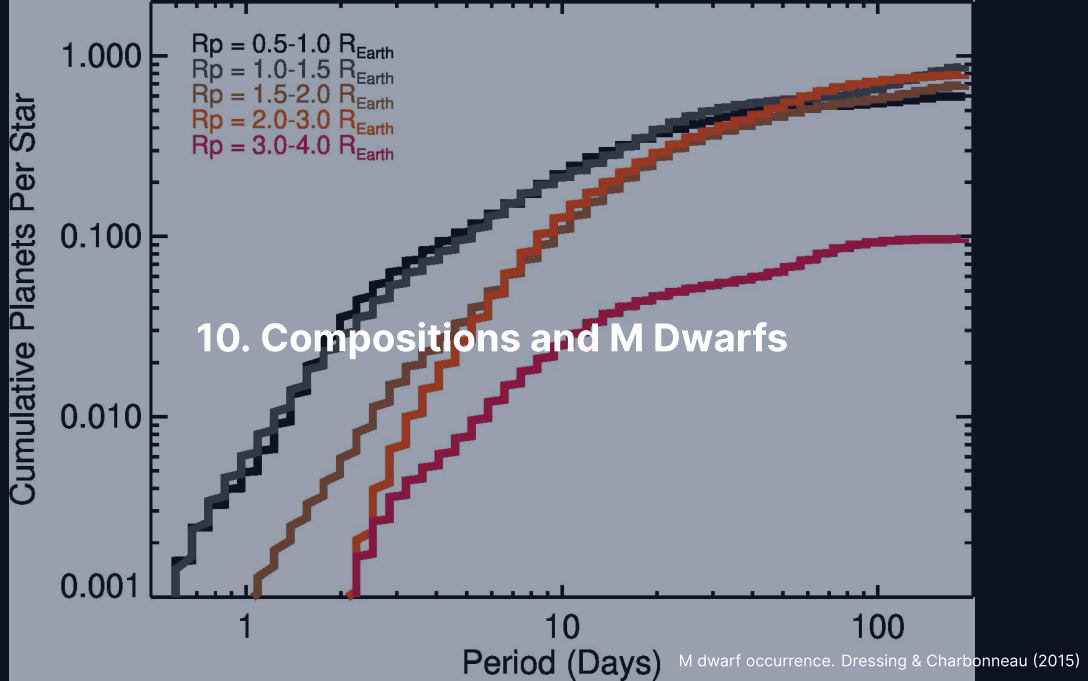
- Transiting planet first occults blueshifted (approaching) hemisphere, then redshifted
- Time-resolved line distortion traces sky-projected spin-orbit angle λ
- Diagnostic tool distinguishing disk migration from dynamical scattering (Triaud 2018)

Observed Obliquity Distribution

Stellar obliquity distributions show that cool stars realign tidally while hot stars retain widely scattered spin-orbit orientations.



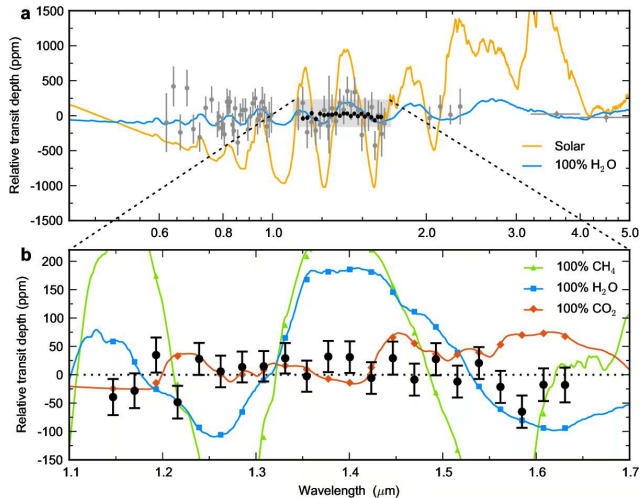
- **Cool stars ($T_{\text{eff}} < 6250 \text{ K}$):** Tight orbits aligned by tidal realignment in convective envelopes
- **Hot stars ($T_{\text{eff}} > 6250 \text{ K}$):** Retain high obliquities due to thin convective zones
- **Primordial scatter:** Wide dispersion at large $a/R_{\star}$ proves dynamical migration occurs



Sub-Neptunes: Water World or H/He?

Sub-Neptune bulk densities suffer a severe degeneracy between extended hydrogen-helium envelopes and deep volatile water layers.

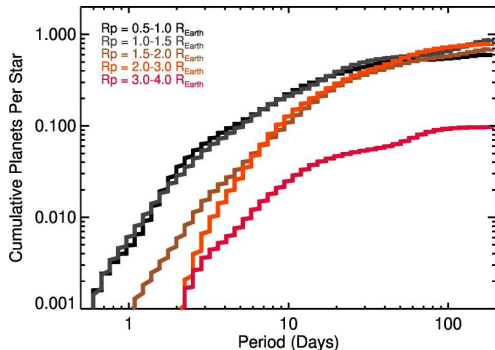
- **Density split:** Super-Earths ($\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$) are rocky; sub-Neptunes ($\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$) require volatile envelopes
- **Degeneracy:** Density cannot separate a 1% H/He envelope from a 50% water-rich mantle
- **Spectroscopic test:** Transmission spectroscopy is required to determine the atmospheric mean molecular weight



Featureless transmission spectrum of the warm sub-Neptune GJ 1214 b from HST data: ruling out cloud-free H_2O , CH_4 , and CO_2 atmospheres demonstrates high-altitude clouds or hazes.

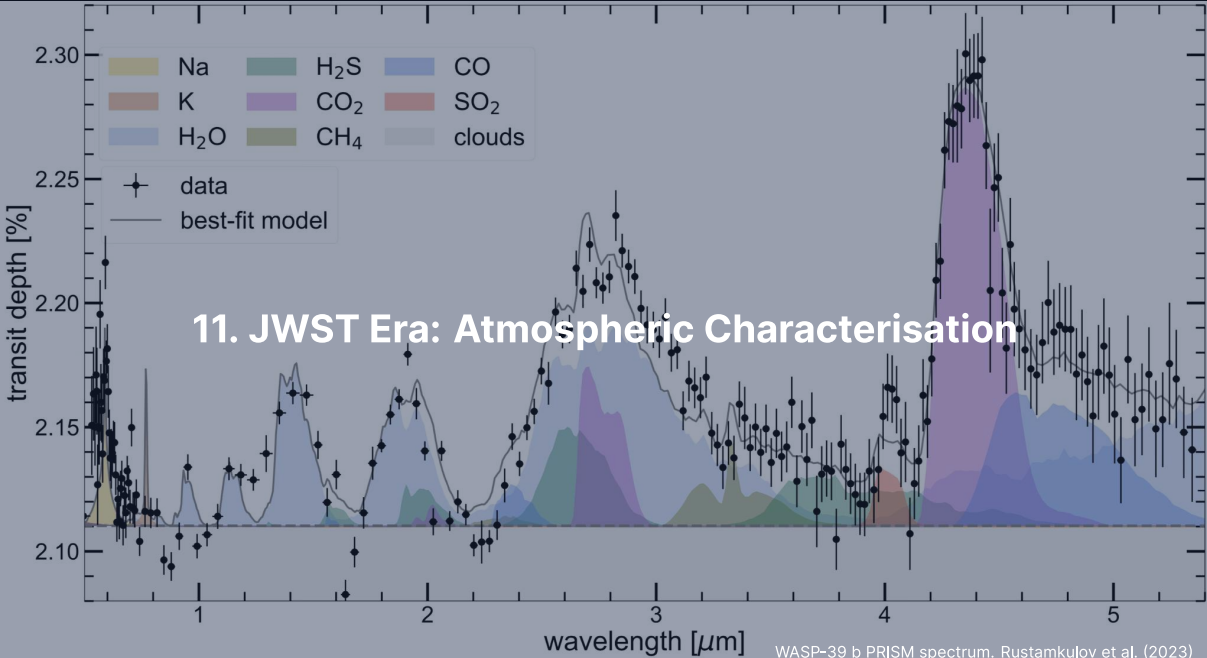
Credit: Kreidberg et al. (2014).

M Dwarfs: Abundant Hosts, Severe Hazards



Credit: Dressing & Charbonneau (2015)

- M dwarfs host ~ 2.5 small planets each; geometrically favourable for transits and RV
- **XUV desiccation:** Extended pre-main-sequence luminosity can strip entire oceans (Luger & Barnes 2015)
- **Tidal locking:** 1:1 spin-orbit resonance forces permanent day and night sides



Transmission Spectroscopy

Starlight filtered through an atmosphere during transit encodes molecular absorption.

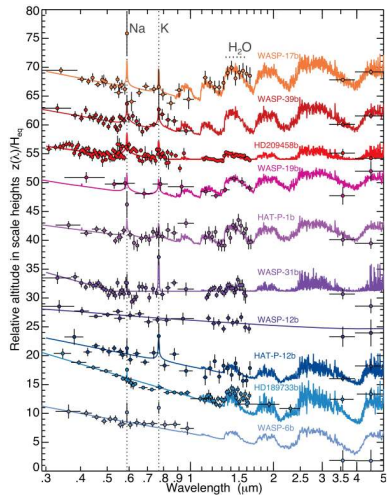
Atmospheric scale height:

$$H = \frac{k_B T}{\mu m_u g}$$

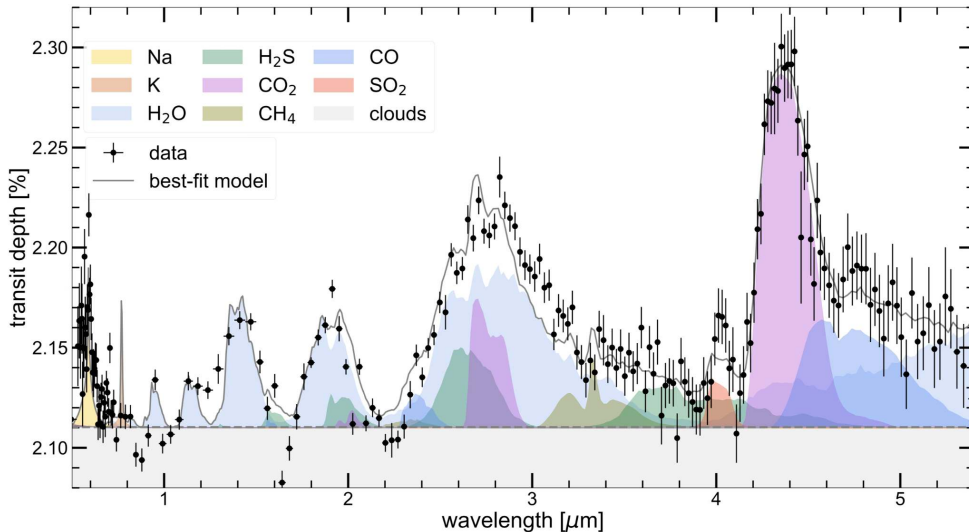
Wavelength-dependent transit depth: $\delta(\lambda) \approx (R_p + n_H(\lambda) H)^2 / R_\star^2$, so $\Delta\delta/\delta \approx 2n_H H / R_p$.

For a hot Jupiter ($T = 1500$ K, $\mu = 2.3$, $g = 25 \text{ m s}^{-2}$):

- Scale height $H \approx 200$ km
- Absolute spectral modulation: ~ 200 ppm (JWST-routine)
- Sub-Neptunes: 10–100 ppm; terrestrial atmospheres: < 10 ppm
- High clouds or hazes flatten the spectrum (GJ 1214 b): a flat spectrum does not mean no atmosphere

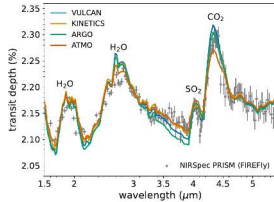


Transmission spectra of ten hot Jupiters from HST and Spitzer across $0.3\text{--}5\ \mu\text{m}$, revealing a continuum from clear atmospheres with prominent water absorption to cloudy, aerosol-dominated scattering slopes. Credit: Sing et al. (2016).

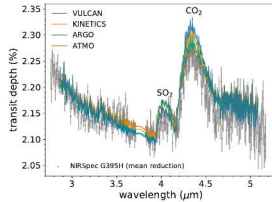


JWST/NIRSpec PRISM transmission spectrum of WASP-39 b. Shaded bands are model opacity contributions; data demonstrate detections of H₂O (33σ), CO₂ (28σ), clouds (21σ), Na (19σ), and CO (7σ). Rustamkulov et al. (2023).

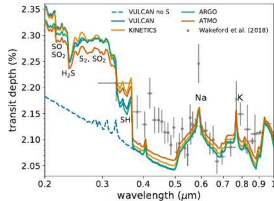
WASP-39 b: First Disequilibrium Photochemistry



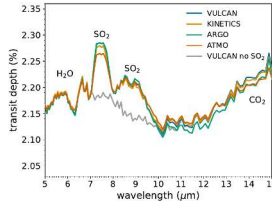
(a)



(b)

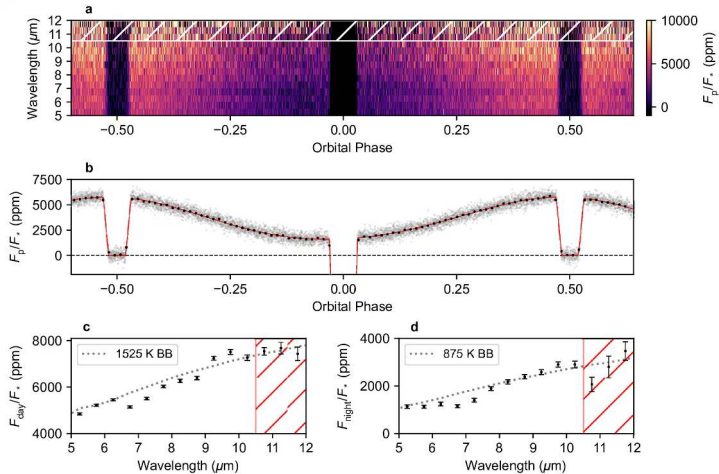


(c)



(d)

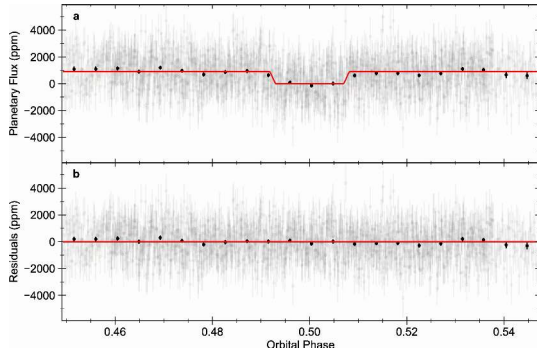
- Hot Saturn ($0.28 M_J$) early release science target
- $4.05 \mu\text{m}$ absorption feature confirms **photochemical** SO_2
- Requires UV photolysis of H_2S , proving disequilibrium chemistry
- First photochemical product identified in an exoplanet atmosphere (Tsai et al. 2023)



JWST/MIRI spectroscopic phase curve of WASP-43 b across orbital phase, showing thermal emission variation from dayside to nightside and constrained atmospheric heat redistribution.

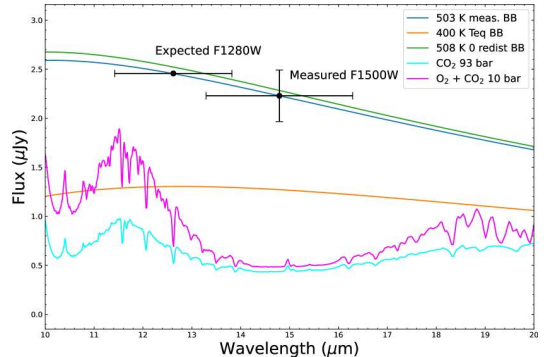
Credit: Bell et al. (2024).

TRAPPIST-1 b: Bare Rock Dayside



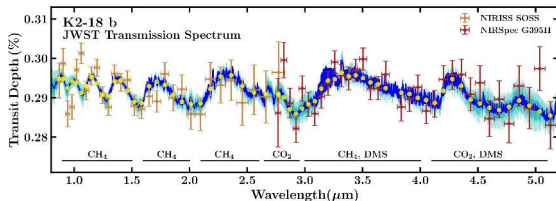
15 μm secondary eclipse

$T_B = 503^{+26}_{-27}$ K, consistent with bare rock equilibrium. **Thick CO₂ atmosphere ruled out**; inner M dwarf rocky planets show a widespread lack of thick atmospheres. Greene et al. (2023).



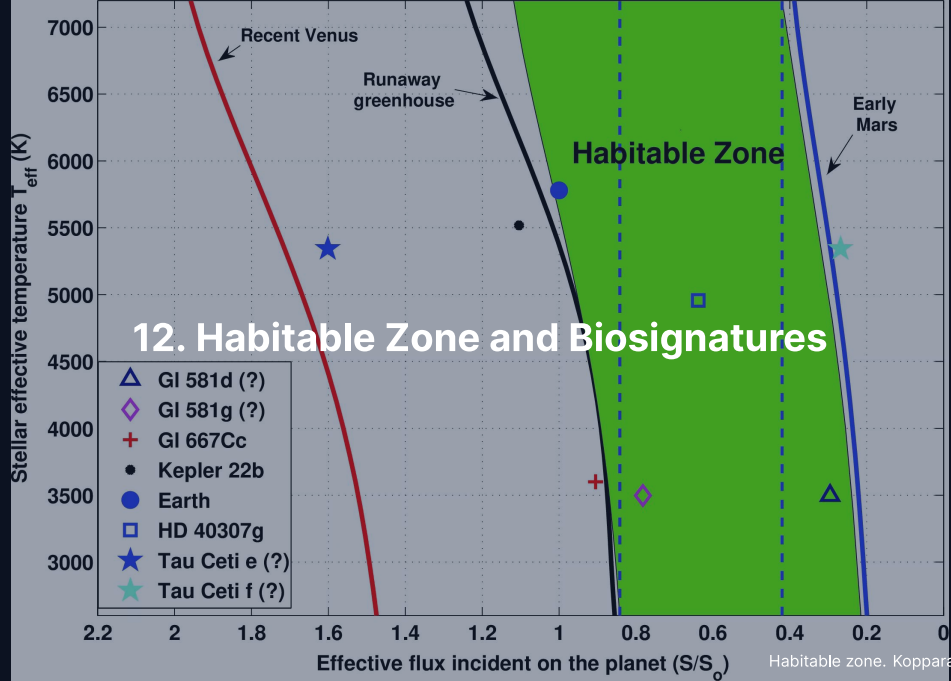
Compared to model atmospheres

K2-18 b: Testing a Tentative Biosignature

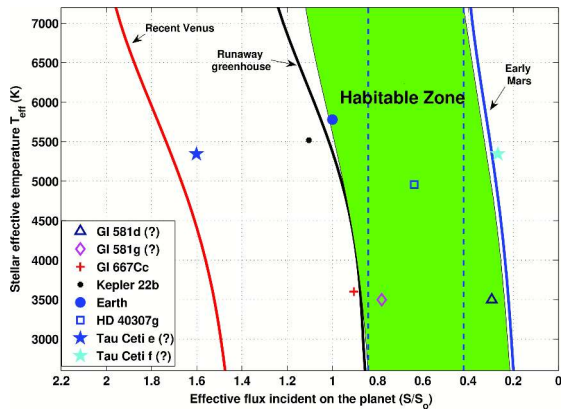


Credit: Madhusudhan et al. (2023)

- Sub-Neptune in M-dwarf HZ with detected CH_4 and CO_2 ; disputed tentative DMS ($\lesssim 2\sigma$)
- Competing models: Hycean ocean, mini-Neptune envelope, or magma ocean
- Exemplifies rigorous real-time vetting required for candidate biosignatures



The Classical Habitable Zone



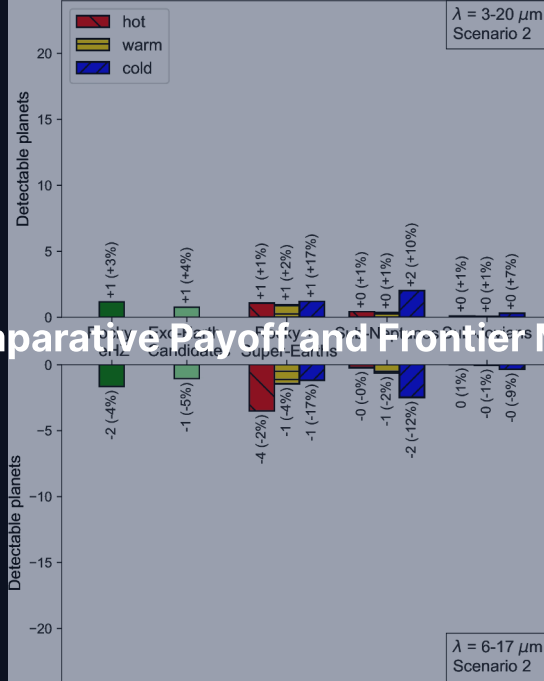
- Inner edge: Simpson-Nakajima runaway greenhouse threshold
- Outer edge: Maximum CO_2 greenhouse before CO_2 condenses
- Boundaries scale with stellar effective temperature and luminosity
- Updated 1D radiative-convective grid from Kopparapu et al. (2013)

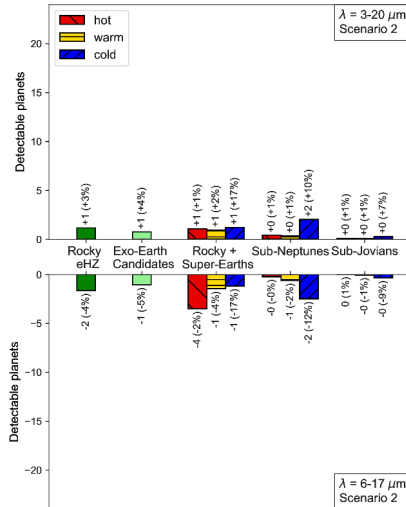
Habitable Zone Evolution and Biosignatures

- **Dynamic trajectory:** HZ boundaries shift with stellar evolution; pre-MS M dwarfs subject planets to prolonged runaway desiccation
- **Disequilibrium pairs:** Atmospheric biosignatures demand chemical disequilibrium ($\text{O}_2 + \text{CH}_4$), not lone gas species
- **False positives:** Abiotic O_2 can build up via H_2O photolysis with hydrogen escape or CO_2 photolysis

Life detection is an inverse problem requiring exclusion of all abiotic pathways.

13. Comparative Payoff and Frontier Missions





Predicted detection yields of the LIFE mid-infrared space interferometer for exoplanets within 20 pc, by radius and insolation. Quanz et al. (2022).

Summary: Today's Course-Level Takeaways

1. **Detection biases:** Transit plus radial velocity breaks $m \sin i$ to yield true bulk density
2. **Demographics:** Small planets dominate ($\eta_{\oplus} \sim 0.4$); photoevaporation carves the radius valley at $\sim 1.8 R_{\oplus}$
3. **Architectures:** resonant chains and peas-in-a-pod systems record smooth disk migration and a shared formation environment
4. **Characterisation:** JWST detects photochemistry and bare rocks; biosignatures require ruling out false positives

Next: Lecture 14



Questions?

Next: Lecture 14. Synthesis & Astrobiology

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience